

Shicheng Fan

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My research explores how AI agents can **recursively improve** their own
memory, harnesses, and self-evolution mechanisms.

First author papers: [ICML 2026](#); [NeurIPS 2026](#); [COLM 2026](#)

EDUCATION

University of Illinois Chicago *Aug 2025 – Present*
Ph.D. in Computer Science. Advisor: **Prof. Philip S. Yu.**

Zhejiang University (*Top 3 in China*) *Sep 2021 – Jun 2025*
B.Eng. in Automation (Control & Robotics), **Chu Kochen Honors College.**
GPA: **3.98/4.0**; Rank: **6/121 (Top 5%)**; **Outstanding Graduate.** Thesis advisor: **Prof. Qi Ye.**

University of Cambridge *Summer 2023*
Short-term Exchange in the AI&ML Summer Program; **Outstanding Student.**

EXPERIENCE

Research Scientist Intern, Aether AI *Aug 2026 – Present*

- **RSIAgent**: autonomously proposes, executes, and verifies exploration tasks, then uses memory to guide the next round. This recursive self-improvement enables open-source GLM-5.3 and Kimi-K3 to surpass GPT-6 Astra in partial-credit scores on OSWorld 2.0 and Agents' Last Exam.
- **Hugging Face Daily Papers #6** (Sep 15, 2026); **Headline feature in Synced (机器之心)**, an AI news outlet.

INVITED TALKS

Kuaishou (Kling AI)	RSIAgent	<i>Sep 2026</i>
Fidelity	Agentic Commerce World	<i>Sep 2026</i>
Visa	Agentic Commerce World	<i>Sep 2026</i>
Alibaba	CoEvoSkills	<i>Jun 2026</i>

PUBLICATIONS & PREPRINTS

First-author Publications

1. [RSIAgent: Autonomous Exploration for Recursive Self-improvement in New Environments.](#)
Sibo Zhu*, **Shicheng Fan***, Xinyue Wang*, Wenyi Wu, Kun Zhou, Biwei Huang. [🔗 Code](#) [🌐 Project Page](#)
ICLR 2027; Under review.
Hugging Face Daily Papers #6 (Sep 15, 2026); **Headline feature in Synced (机器之心)**, an AI news outlet.
TL;DR Agents autonomously propose tasks, act, verify outcomes, and retain reusable experience in memory to guide broad-then-deep exploration. This RSI loop lets open-source GLM-5.3 and Kimi-K3 surpass GPT-6 Astra in partial-credit scores on OSWorld 2.0 (78.98 vs. 72.60) and Agents' Last Exam (84.82 vs. 82.26).
2. [Agentic Commerce World: An Auditable and Verifiable Environment for Vibe Commerce.](#)
Shicheng Fan, Mingdai Yang, Duohao Wang, Canyu Chen, Yongfeng Zhang, Hua Wei, Manling Li, Julian McAuley, Kun Zhang, Philip S. Yu, Kejing Yu, Zhiwei Liu. [🔗 Code](#) [🌐 Project Page](#)
KDD 2027; Under review.
TL;DR Provides a reproducible testbed with auditable transaction traces for buyer and merchant agents. Evaluation on 200 capability tasks and 60 large-catalog tasks covering 785K listings reveals failures missed by final-state metrics.
3. [CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification.](#)
Hanrong Zhang*, **Shicheng Fan***, Henry Peng Zou, Yankai Chen, Zhenting Wang, Jiayu Zhou, Chengze Li, Wei-Chieh Huang, Yifei Yao, Kening Zheng, Xue Liu, Xiaoxiao Li, Philip S. Yu. [🔗 Code](#) [🌐 Project Page](#)
COLM 2026; Accepted.
TL;DR Evolves executable skills with a Skill Generator and a co-evolving Surrogate Verifier, without access to ground-truth tests. Skills beat five SkillsBench baselines on Claude Code and Codex and transfer to six other LLMs.

4. **TRACE: Trajectory Recovery for Continuous Mechanism Evolution in Causal Representation Learning.**
Shicheng Fan, Kun Zhang, Lu Cheng.  [Code](#)
ICML 2026; Accepted.
TL;DR Models evolving causal mechanisms as mixtures of atomic ones, with identifiability for latent variables and mixing trajectories. Experiments reach up to 0.99 trajectory correlation and outperform discrete-switching baselines.
5. **MOSAIC: Module Discovery via Sparse Additive Identifiable Causal Learning for Scientific Time Series.**
Shicheng Fan, Nour Elhendawy, Jianle Sun, Ke Fang, Kun Zhang, Yihang Wang, Lu Cheng.  [Code](#)
NeurIPS 2026; Accepted.
TL;DR Maps latent variables to interpretable observation modules using identifiable temporal causal learning and sparse support recovery. Results on RNA dynamics and tokamak simulations agree with domain knowledge.
6. **Beyond Math and Code: Lightweight Corpus-Grounded Process Rewards for Factual Question Answering.**
Shicheng Fan*, Haochang Hao*, Dehai Min*, Weihao Liu, Hanrong Zhang, Lingwei Wei, Henry Peng Zou, Chengquan Guo, Jie Yang, Honghui Bao, Zhiwei Liu, Lu Cheng, Philip S. Yu.  [Code](#)
ICLR 2027; Under review.
TL;DR Corpus co-occurrence yields sentence-level rewards that improve all 30 factual-QA settings and beat four RL baselines in 17 of 20 comparisons. Full training is 5.5–10.4× faster, averaged over Qwen3-4B and Qwen3-8B.

Collaborative Publications

1. **Paying for Honesty Without Knowing the Truth: Reputation-Penalty Design for LLM Marketplace Agents.**
Mingdai Yang, Shicheng Fan, Kejing Yu, Duohao Wang, Li Sun, Hao Peng, Philip S. Yu, Zhiwei Liu.
KDD 2027; Under review.
TL;DR Uses noisy complaints to penalize deceptive LLM merchants without product-level truth. With SPARC, CARP recovers most of the welfare gap to a perfect-information oracle and achieves the best welfare among tested policies.
2. **Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey.**
Hanrong Zhang, Yankai Chen, Shicheng Fan, Dehai Min, Shaowen Chen, . . . , Dawn Song, Philip S. Yu, Xue Liu.
Preprint, June 2026.
TL;DR Offers a lifecycle view of synthetic data for LLM agents, from task grounding to downstream evaluation. The survey pinpoints the reliability barriers that make agent learning difficult to scale.